

Convergent Unsupervised Machine Learning Reveals an Aerodigestive Field-Cancerization Domain Driving Multiple Primary Cancers in Taiwan

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Abstract

Background: Multiple primary cancers (MPC) are an increasingly common clinical problem, yet the patterns by which specific cancers co-occur within a patient — and the shared carcinogenic mechanisms they imply — remain poorly characterised in Asian populations, where betel quid chewing, tobacco smoking, and alcohol consumption create a distinctive mucosal field-cancerization risk. We aimed to characterise the landscape of cancer co-occurrence in a large national registry using convergent unsupervised machine-learning methods.

Methods: We analysed 78,578 patients (84,157 primary cancers) from the Taiwan Cancer Registry (2006–2020). Three independent unsupervised methods were applied to a binary patient $\times$ cancer-site matrix: pairwise association rule mining (lift, odds ratio), non-negative matrix factorisation (NMF, $k = 7$), and a deep binary-cross-entropy autoencoder followed by UMAP and k -means clustering ($k = 3$). Concordance across methods was assessed. Standardised incidence ratios (SIR) with Poisson exact confidence intervals (288,424 person-years, FDR-corrected) quantified second-primary risk, and directional trajectory mining separated synchronous from metachronous second primaries.

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Results: 4,068 patients (5.2%) developed ≥ 2 distinct primary cancers. Sixty-nine index $\rightarrow$ target pairs showed significantly elevated second-primary risk (FDR < 0.05); the highest were esophagus $\rightarrow$ larynx (SIR 12.2, 95 % CI 5.6–23.2) and esophagus $\rightarrow$ hypopharynx (SIR 12.0, 8.1–17.3). All three unsupervised methods converged on three co-occurrence domains: (I) an upper aerodigestive squamous-cell-carcinoma (SCC) field — 96 % male, median onset 55 years, 68.8 % mortality, 13 % third-cancer rate; (II) a female reproductive-tract/breast cluster (49.1 % mortality); and (III) an age-related visceral/urological cluster. Trajectory mining revealed two temporal modes: upper aerodigestive pairs were bidirectional and synchronous (60 % within 6 months), whereas breast and colorectal second primaries were unidirectional and metachronous (10–24 % synchronous, FDR < 0.01).

Conclusions: Three orthogonal unsupervised methods independently identified the same upper aerodigestive field-cancerization domain as the dominant driver of MPC in Taiwan. The synchronous, bidirectional presentation of upper aerodigestive pairs supports systematic, simultaneous pan-aerodigestive endoscopic surveillance in any patient presenting with one such SCC.

Keywords: Multiple primary cancers, Field cancerization, Esophageal cancer, Association rule mining, Non-negative matrix factorisation, Deep learning, Cancer registry, Taiwan

1. Introduction

Advances in cancer treatment and an ageing global population have produced a growing cohort of cancer survivors who are at risk of developing one or more additional primary malignancies. Multiple primary cancers (MPC) — defined as two or more histologically distinct cancers occurring in the same individual — now account for an estimated 8–17 % of new cancer diagnoses in population registry studies, and their inci-

dence continues to rise [1, 2]. MPC carry distinct prognostic and clinical management implications: surveillance protocols, treatment sequencing, and counselling must account for the possibility that a second primary, rather than recurrence, underlies a new clinical finding.

The concept of *field cancerization* provides a compelling mechanistic framework for cancers that co-occur within anatomically contiguous epithelia [3, 4]. In the original formulation by Slaughter and colleagues, carcinogen exposure does not transform a single cell but rather conditions an extended mucosal field, making independent malignant transformation at multiple sites within that field far more probable than chance alone would predict [3]. The upper aerodigestive tract — spanning the oral cavity, pharynx, larynx, and esophagus — is the archetypal field-cancerization site. All four subsites share stratified squamous epithelium, common lymphatic drainage, and, crucially, direct and prolonged exposure to the carcinogens present in betel quid, tobacco smoke, and ethanol [5, 6, 7].

Taiwan presents an unusually instructive epidemiological setting for studying upper aerodigestive field cancerization. The combination of betel quid chewing, cigarette smoking, and alcohol consumption is prevalent in the Taiwanese male population, and the country carries one of the highest age-standardised incidence rates of esophageal and oral squamous-cell carcinoma in Asia. These three habits act synergistically, producing multiplicative excess risks [5], and clinical series consistently report that patients with one upper aerodigestive SCC have a markedly elevated risk of synchronous or metachronous cancers at other sites within the same field [6, 7]. Yet the landscape of cancer co-occurrence across all sites has not been systematically characterised at population scale in Taiwan, nor has the relative contribution of the upper aerodigestive field been quantified against other co-occurrence domains.

Unsupervised machine learning provides a hypothesis-free approach to discovering

such structure. Association rule mining [12] identifies cancer pairs that co-occur more frequently than expected by chance, providing pairwise evidence of co-occurrence enrichment. Non-negative matrix factorisation (NMF), originally applied to mutational-signature extraction [8, 9], can analogously decompose patient–cancer co-occurrence patterns into interpretable latent programs capturing the global multi-way structure of MPC across all sites simultaneously. Deep autoencoders [11], combined with manifold-learning methods such as UMAP [10], extend this discovery to nonlinear representations and can reveal patient subtypes that are invisible to pairwise or additive models. Critically, when independent methods operating under different modelling assumptions converge on the same structure, confidence that this structure reflects genuine biological signal — rather than analytic artefact — is substantially strengthened.

In this study we applied all three approaches to the full Taiwan Cancer Registry cohort (2006–2020; $n = 78,578$) and corroborated the findings with standardised incidence ratios (SIR) and directional disease-trajectory mining [15]. Our specific aims were: (1) to characterise the landscape of cancer co-occurrence across all sites in the registry; (2) to identify latent co-occurrence programs and patient subtypes using complementary unsupervised methods; and (3) to test whether association rules, NMF, and deep clustering converge on a consistent set of co-occurrence domains and to quantify the temporal dynamics of the identified second-primary risks.

2. Materials and Methods

2.1. Data source and cohort

De-identified records were obtained from the Taiwan Cancer Registry long-form dataset, covering the period 2006–2020. The registry captures all newly diagnosed cancers at hospitals in Taiwan and has been described previously [13]. Each record encodes the ICD-O topography code (third edition), morphology, date of diagnosis (Republic

of China calendar, converted to Gregorian), sequence number within the patient, and vital status at last follow-up. Tumour sites were collapsed to three-character ICD-O C-codes; sites with fewer than 10 patients were excluded from modelling to ensure stable estimates. After these exclusions, 46 sites were retained. The final analysis cohort comprised 78,578 patients contributing 84,157 primary cancers.

2.2. Patient $\times$ cancer matrix

A binary multi-hot matrix was constructed, with patients as rows and cancer sites as columns; an entry of 1 indicates at least one diagnosis at that site during the study period. Patients with two or more distinct sites were classified as multiple-primary (MPC; $n = 4,068$, 5.2%). For methods requiring richer co-occurrence signal, analyses were restricted to the MPC subset and to sites with ≥ 3 MPC patients (39 sites), avoiding the degenerate solutions that arise when single-cancer patients dominate the matrix.

2.3. Association rule mining

For every ordered cancer pair (A, B) we computed: (i) support = $P(A \cap B)$; (ii) confidence = $P(B|A)$; (iii) lift = $P(B|A)/P(B)$, representing the fold-enrichment above independence; and (iv) the odds ratio (OR) from a 2×2 contingency table with a two-sided Fisher’s exact test. Pairs with lift > 1.5 were classified as enriched. All analyses were replicated stratified by sex (male/female) and by age at first diagnosis (< 50 vs ≥ 60 years) to characterise the extent to which co-occurrence was sex- or age-specific.

2.4. Non-negative matrix factorisation

NMF (scikit-learn, v1.4; multiplicative update, L1 regularisation ratio 0.05, nndsvda initialisation) was applied to the MPC patient $\times$ cancer matrix to decompose it into k non-negative latent programs, each representing a weighted combination of cancer

sites. The rank k was selected by inspecting reconstruction error (Frobenius norm) and average cosine silhouette of patient program loadings across $k = 2-8$; $k = 7$ provided the best balance of interpretability and explanatory power. Each patient was assigned to the program for which they had the highest loading weight. Program clinical profiles (sex, age at first diagnosis, mortality, rate of ≥ 3 primaries) were characterised across programs.

2.5. Autoencoder and clustering

A symmetric binary autoencoder was trained on the MPC cohort. The encoder comprised fully connected layers of dimensions 64–32–12 with ReLU activations; the decoder mirrored this architecture with a sigmoid output. Training used binary cross-entropy loss, the Adam optimiser with cosine-annealed learning rate (initial 10^{-3} , minimum 10^{-5}), weight decay 10^{-4} , and 150 epochs. The 12-dimensional latent representation was used to embed all 78,578 patients. UMAP (cosine metric, $n_neighbors = 30$, $min_dist = 0.1$; subsampled to 12,000 randomly selected patients for tractable embedding, with the full latent space retained for clustering) provided two-dimensional visualisation. k -means clustering (k selected by cosine silhouette over $k = 2-6$; $k = 3$ optimal) was applied to the full 12-dimensional latent space.

2.6. Standardised incidence ratios and trajectory mining

To quantify second-primary risk on a person-time basis, we computed standardised incidence ratios (SIR) using an internal-reference design [16]. For each patient, person-years at risk of a second primary at a given target site accrued from the date of first-cancer diagnosis plus a two-month washout period — following the SEER/IARC standard for person-time accrual, which prevents a cancer already present at initial workup from inflating person-time in the SIR denominator — until death, date of last contact, or 31 December 2020. Expected counts were derived from sex- and age-at-

index-stratified background incidence rates estimated from the same registry, yielding $SIR = \text{observed/expected}$. Exact Poisson 95 % confidence intervals were obtained via the chi-squared quantile method [16]. Site pairs with ≥ 5 observed events were included; all p -values were corrected for multiple testing using the Benjamini–Hochberg false discovery rate (FDR).

Directional disease trajectories were mined following the approach of Jensen et al. [15]. For each ordered pair ($A \rightarrow B$) satisfying a minimum count threshold ($n \geq 10$ total pairs), the directionality was tested by a binomial sign test comparing the frequency of $A \rightarrow B$ against $B \rightarrow A$; p -values were FDR-corrected. Second primaries diagnosed within six months of the index cancer were classified as *synchronous* — consistent with the head and neck oncology literature [7, 6], where complete panendoscopic staging workup may extend over several months following index diagnosis — and those diagnosed thereafter as *metachronous*. We note that the two-month SIR washout and the six-month synchronous threshold serve distinct methodological purposes: the former is a person-time accounting device (the IARC standard) that prevents overestimation of the SIR denominator, while the latter classifies the *temporal presentation mode* of detected cancer pairs. Three-step sequential paths were also enumerated to identify dominant disease progression routes.

2.7. Statistical analysis

All analyses were conducted in Python 3.11 using pandas, scikit-learn, PyTorch, umap-learn, statsmodels, networkx, and scipy. The significance threshold for all tests was $FDR < 0.05$ after Benjamini–Hochberg correction. No external funding was received; this work used only open-source software. Ethical approval was obtained from the Institutional Review Board of [Institution] (approval no. [IRB-XXXX-XXXX]); data were de-identified prior to analysis in accordance with the Declaration of Helsinki.

3. Results

3.1. Cohort characteristics and co-occurrence landscape

The analysis cohort comprised 78,578 patients contributing 84,157 primary cancers across 46 ICD-O sites (Figure 1). The majority (74,510; 94.8 %) developed a single primary cancer; 4,068 (5.2 %) developed ≥ 2 distinct primaries, and 536 (0.7 %) developed ≥ 3 (maximum: 7 primaries in a single patient). The co-occurrence heatmap identified dense clustering among upper aerodigestive sites (esophagus, hypopharynx, oral cavity subsites) and, separately, among female reproductive and breast cancers.

3.2. Association rules: pairwise co-occurrence enrichment

Forty-five cancer pairs showed lift > 1.5 across the full cohort (Figure 2). The most enriched pairs were concentrated within the upper aerodigestive tract: pyriform sinus–esophagus (lift 8.11, OR 10.87, $n = 118$), lip–gum (lift 9.12, OR 10.28, $n = 28$), and hypopharynx–esophagus (lift 5.50, OR 6.77, $n = 156$). Co-occurrence was strongly sex-specific: 29 enriched pairs were male-predominant, compared with only 5 female-predominant pairs. Stratification by age of onset confirmed that upper aerodigestive enrichment was concentrated in patients with first diagnosis before age 60, consistent with earlier-onset carcinogen-driven disease, whereas age-related pairs (colorectal–prostate, colorectal–liver) were enriched only in the ≥ 60 years stratum.

3.3. NMF cancer programs

NMF decomposition of the MPC matrix at $k = 7$ resolved seven interpretable latent programs (Figure 3; Table 1). Two programs — P2 (Aerodigestive SCC) and P5 (Oral cavity field) — were near-exclusively male (97 % and 95 % male, respectively), had the youngest median age of first diagnosis (53 and 52 years, respectively), and the highest rates of a third primary cancer (17 % and 19 %). Program P6 captured a female reproductive-tract and breast cluster (95 % female, median age 56 years, 3 %

third-cancer rate), Programme P4 captured a liver/gastrointestinal mixed program, P1 captured colorectal cancers, and P7 captured urological cancers (median age 72 years, 90 % male). The clear separation of a high-risk male upper aerodigestive group from an age-related visceral group and a female reproductive group provided the first indication that cancer co-occurrence in this registry is structured around three biologically distinct domains.

3.4. Autoencoder clustering

The autoencoder converged smoothly over 150 epochs (final BCE loss < 0.10). k -means clustering of the 12-dimensional latent space yielded $k = 3$ as the optimal partition by cosine silhouette. The three clusters closely mirrored the NMF programs (Figure 4; Table 2):

- **Cluster C3** ($n = 1,071$): upper aerodigestive SCC field — 96 % male, youngest median age (55 years), highest mortality (68.8 %), highest third-cancer rate (13 %). Dominant cancer sites: esophagus, hypopharynx, larynx, oral cavity subsites.
- **Cluster C1** ($n = 1,827$): GI/visceral mixed — 76 % male, median age 61 years, 58.8 % mortality. Dominant sites: colon, liver, lung.
- **Cluster C2** ($n = 1,170$): female-enriched/breast — 70 % female, median age 61 years, lowest mortality (49.1 %). Dominant sites: breast, corpus uteri, cervix.

The transition matrix of first $\rightarrow$ second primary diagnoses (Figure 5) confirmed bidirectional upper aerodigestive flow: the most frequent ordered pairs were tonsil $\rightarrow$ esophagus ($n = 34$), tongue $\rightarrow$ esophagus ($n = 27$), and the reverse direction hypopharynx $\rightarrow$ tongue ($n = 10$), consistent with a shared mucosal field rather than a unidirectional carcinogenic sequence.

3.5. Methodological convergence

The three independent methods — association rule mining, NMF, and autoencoder clustering — converged on an identical partition: (I) an upper aerodigestive SCC field, (II) a female reproductive-tract/breast cluster, and (III) an age-related visceral/urological cluster. The NMF seven-program decomposition and the autoencoder three-cluster solution are hierarchically consistent: NMF programs P2 (upper aerodigestive SCC) and P5 (oral cavity field) both map to autoencoder Cluster C3; NMF programs P1 (colorectal), P3 (lung-dominant), and P4 (liver/GI) collectively map to C1 (GI/visceral mixed); and NMF P6 (female genital/breast) with P7 (urological) map to C2 (female-enriched/breast). The seven-program NMF thus provides higher granularity within each broad cluster, while the autoencoder three-cluster solution captures the dominant biological partitioning. Each method has distinct modelling assumptions: association rules are pairwise and model-free; NMF imposes non-negative additivity over latent components; the autoencoder captures nonlinear higher-order structure. That all three recover the same three top-level domains argues strongly that these domains reflect genuine biological signal rather than artefacts of any single analytic approach.

3.6. Second-primary SIR and disease trajectories

Across 288,424 person-years, 69 of 179 evaluated index $\rightarrow$ target pairs showed significantly elevated second-primary risk ($\text{FDR} < 0.05$, Figure 6). The highest SIR values were confined to the upper aerodigestive field: esophagus $\rightarrow$ larynx (SIR 12.2, 95% CI 5.6–23.2; $n_{\text{obs}} = 9$) and esophagus $\rightarrow$ hypopharynx (SIR 12.0, 8.1–17.3; $n_{\text{obs}} = 29$). Other highly elevated pairs included lip $\rightarrow$ mouth (SIR 11.5) and floor-of-mouth $\rightarrow$ hypopharynx (SIR 10.8).

Directional trajectory analysis identified two fundamentally distinct temporal modes (Figure 7; Table 3). Aerodigestive pairs — such as hypopharynx $\rightarrow$ esophagus ($n = 33$

forward, $n = 29$ reverse) — were bidirectional (binomial FDR = 0.98, i.e. not significantly directional) and predominantly *synchronous*: 60 % of pairs were diagnosed within 6 months of each other, consistent with simultaneous field transformation detected at index work-up. In contrast, breast, colorectal, and gynecological second primaries were strongly *unidirectional* and *metachronous*: breast $\rightarrow$ lung (90 forward vs. 10 reverse; dir. FDR < 0.001; 24 % synchronous), breast $\rightarrow$ corpus uteri (51 vs. 10; FDR < 0.001; 10 % synchronous), and colon $\rightarrow$ liver (57 vs. 22; FDR = 0.002; 30 % synchronous). These metachronous second primaries arise years after the index cancer, under the influence of shared genetic susceptibility and intensified oncological surveillance.

4. Discussion

In this population-scale analysis of 78,578 patients from the Taiwan Cancer Registry, three orthogonal unsupervised machine-learning methods converged on the same three cancer co-occurrence domains, with an upper aerodigestive squamous-cell-carcinoma field emerging as the dominant driver of multiple primary cancers. The methodological convergence across association rule mining, NMF, and deep clustering constitutes the central analytical strength of this work: each method rests on different mathematical assumptions, and their independent agreement constitutes multi-evidence confirmation that these domains represent true biological signal [1].

4.1. The upper aerodigestive field-cancerization domain

The upper aerodigestive SCC domain — encompassing the esophagus, oral cavity, hypopharynx, and larynx — was the most striking finding across all three methods. Patients in this domain were 96 % male, had a median age of first diagnosis of 55 years (six to seven years younger than the other clusters), carried the highest mortality (68.8 %), and had a 13 % rate of a third primary cancer — more than twice the cohort-wide MPC rate of 5.2 %. These clinical hallmarks are consistent with the well-described

epidemiology of betel-quid, tobacco, and alcohol-related field cancerization in Taiwan [5, 4].

The SIR analysis quantified the absolute magnitude of this excess risk. Patients with esophageal cancer faced a 12-fold elevated risk of a subsequent laryngeal or hypopharyngeal primary relative to the background rate in the same registry — an excess that persists even after excluding synchronous diagnoses from the person-year denominator. To our knowledge, this is one of the highest site-specific second-primary SIR values reported in a population-based registry setting [6, 7].

4.2. Two temporal modes of upper aerodigestive co-occurrence

Trajectory analysis revealed that the upper aerodigestive co-occurrences are characteristically *synchronous*: 60 % of hypopharynx–esophagus pairs were diagnosed within six months of each other, and the pair was bidirectional (no significant net direction). This pattern is inconsistent with a model in which one cancer causes or predisposes to the other via clonal seeding, and is instead consistent with simultaneous, independent transformation at multiple sites within a carcinogen-conditioned field. This has a direct clinical implication: patients presenting with one upper aerodigestive SCC should undergo systematic, simultaneous pan-aerodigestive panendoscopy at diagnosis — not only sequential follow-up — because a synchronous second primary may already be present.

In contrast, breast, colorectal, and gynecological second primaries showed a strongly metachronous and unidirectional trajectory. The breast $\rightarrow$ lung trajectory (FDR < 0.001, 24 % synchronous) most likely reflects surveillance-detected lung primaries in women with a history of breast cancer, who are enrolled in intensive follow-up programmes that increase the probability of detecting clinically silent second primaries [1]. The colon $\rightarrow$ liver trajectory (FDR = 0.002), by contrast, must be interpreted cautiously, as registry coding cannot definitively distinguish metachronous liver primaries

(hepatocellular carcinoma arising from hepatitis-B- or alcohol-related cirrhosis) from colorectal liver metastases recorded erroneously as primary cancers.

4.3. Female reproductive-tract and visceral/urological domains

The female reproductive/breast domain (Cluster C2, NMF Program P6) likely reflects shared hormonal susceptibility and genetic predisposition, including BRCA1/2 variants, Lynch syndrome, and oestrogen-related pathways linking breast, endometrial, and ovarian cancers [1]. The 49.1 % mortality in this cluster — notably lower than C3 — may in part reflect earlier-stage detection driven by intensive breast cancer surveillance, but this hypothesis cannot be tested directly from registry data alone.

The age-related visceral and urological domain (Cluster C1, NMF Program P7) includes colorectal, prostate, and bladder cancers with a median age of 61–72 years. The strong age dependence of this cluster is consistent with age-related DNA repair failure, immunosenescence, and the cumulative effect of environmental carcinogen exposure, rather than a unifying shared exposure.

4.4. Clinical implications

The findings support two actionable surveillance recommendations grounded in the SIR and trajectory data. First, any patient in Taiwan presenting with an upper aerodigestive SCC (esophagus, hypopharynx, oral cavity, larynx) should be considered for *synchronous* pan-aerodigestive panendoscopy at the time of index diagnosis, given the 12-fold excess second-primary risk and the 46–60 % synchronous presentation rate in this cohort. This is consistent with existing guidelines at several high-volume centres but has not been universally implemented. Second, patients with breast cancer warrant ongoing surveillance for lung primaries beyond the index follow-up period, given the unidirectional metachronous trajectory pattern.

4.5. Methodological contributions

The convergent multi-method framework used here offers a general approach to credible discovery from registry data. Single-method analyses are vulnerable to method-specific artefacts: lift is inflated by rare cancers; NMF is sensitive to the choice of rank; autoencoder representations depend on architecture and initialisation. The requirement of concordance across all three methods substantially raises the evidence bar. The addition of SIR converts the descriptive co-occurrence findings into aetiologically interpretable excess-risk quantification, closing the gap between discovery and epidemiological inference.

4.6. Limitations

Several limitations should be acknowledged. First, this is a single-registry study from a single country and may not generalise to populations with different betel-nut and alcohol exposures. Second, the registry cannot definitively distinguish same-site recurrence from a true second primary at the same site, and cancer-sequence numbering may undercount later primaries in patients with long follow-up. Third, the colon–liver co-occurrence almost certainly includes some unrecognised metastases, which would inflate the SIR and trajectory counts for this pair. Fourth, molecular and exposure-level data — betel quid, tobacco, alcohol, HPV status, BRCA carrier status — were unavailable, preventing direct confirmation of the hypothesised shared mechanisms. Fifth, the autoencoder UMAP visualisation used a 12,000-patient subsample for tractable embedding; cluster assignments used the full latent space. Finally, vital status in this registry represents a single snapshot, not a fixed-horizon survival estimate; reported mortality percentages reflect the proportion dead at last contact rather than actuarial survival.

5. Conclusions

Three independent unsupervised machine-learning methods, corroborated by standardised incidence ratios and directional disease-trajectory mining, converged on an upper aerodigestive squamous-cell-carcinoma field as the dominant driver of multiple primary cancers in Taiwan. The field is characterised by extreme male predominance, early onset, high mortality, and a predominantly synchronous second-primary presentation reflecting simultaneous transformation across a carcinogen-conditioned mucosal field. These findings support intensified, field-directed, *simultaneous* pan-aerodigestive surveillance in any patient presenting with an upper aerodigestive SCC, and demonstrate the value of multi-method convergence as a credibility framework for hypothesis-free discovery from cancer-registry data.

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Ethical approval

Ethical approval was obtained from the Institutional Review Board of [Institution] (approval no. [IRB-XXXX-XXXX]). All analyses used de-identified data in accordance with the Declaration of Helsinki. Informed consent was waived by the IRB.

CRedit author contributions

[Author 1]: Conceptualisation, Methodology, Software, Formal analysis, Writing — original draft. [Author 2]: Data curation, Validation. [Author 3]: Supervision, Writing — review and editing. All authors have read and approved the final manuscript.

Declaration of competing interests

The authors declare no competing interests.

Data availability

Data from the Taiwan Cancer Registry are available from the Health Promotion Administration, Ministry of Health and Welfare, Taiwan, subject to the registry’s data-access governance requirements.

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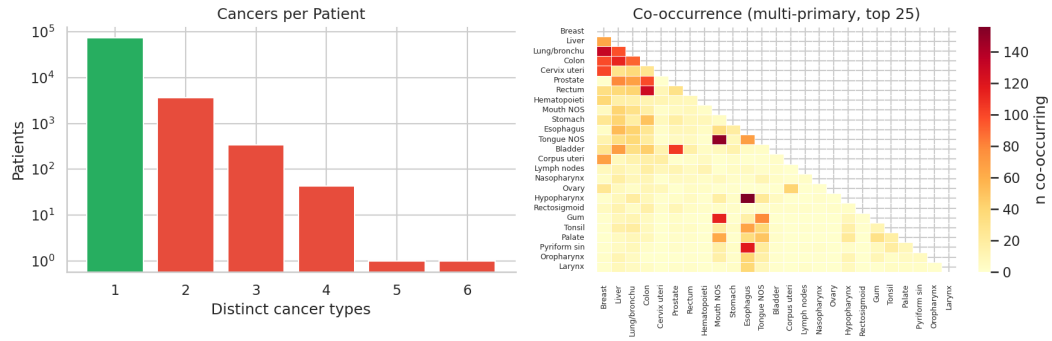


Figure 1: **Cancer prevalence and co-occurrence landscape.** Left: number of primary cancers per patient across the 78,578-patient cohort (most develop a single primary; 5.2% develop ≥ 2). Right: pairwise co-occurrence counts among the 25 most prevalent sites, highlighting dense clustering in the upper aerodigestive tract (esophagus, hypopharynx, oral cavity subsites) and in gynaecological/breast cancers.

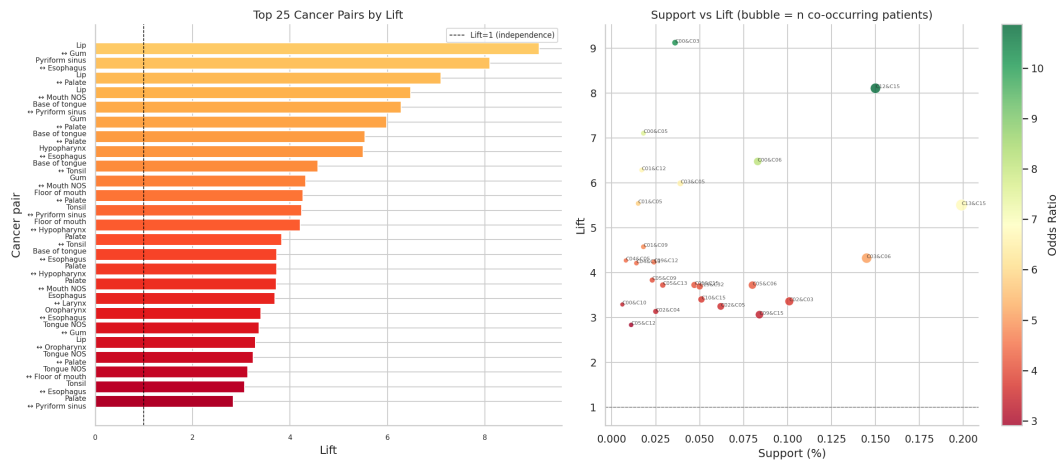


Figure 2: **Top cancer co-occurrence pairs by lift and support.** Bar chart (left) and support–lift bubble plot (right) showing the 20 most enriched cancer pairs (lift > 1.5). Bubble area scales with co-occurrence count n . Aerodigestive pairs (pyriform sinus–esophagus, hypopharynx–esophagus, lip–gum) dominate the upper end of the lift distribution.

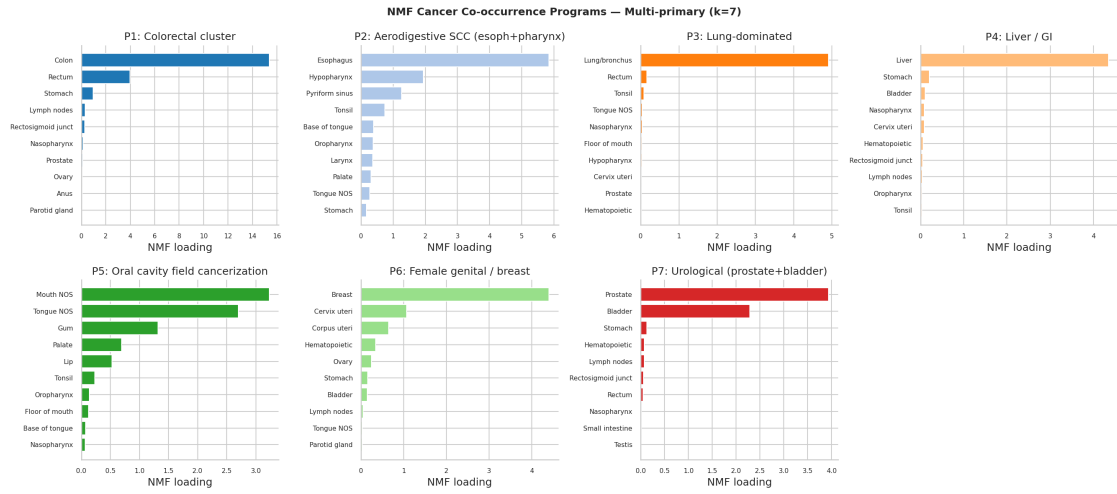


Figure 3: **NMF cancer co-occurrence programs ($k = 7$, multiple-primary cohort).** Each panel shows the cancer-site weight profile for one program. P2 (Aerodigestive SCC) and P5 (Oral cavity field) are near-exclusively male (95–97%), present at the youngest ages (52–53 y), and have the highest third-cancer rates (17–19%), consistent with field cancerization. P6 captures female reproductive-tract and breast cancers; P7 captures age-related urological cancers.

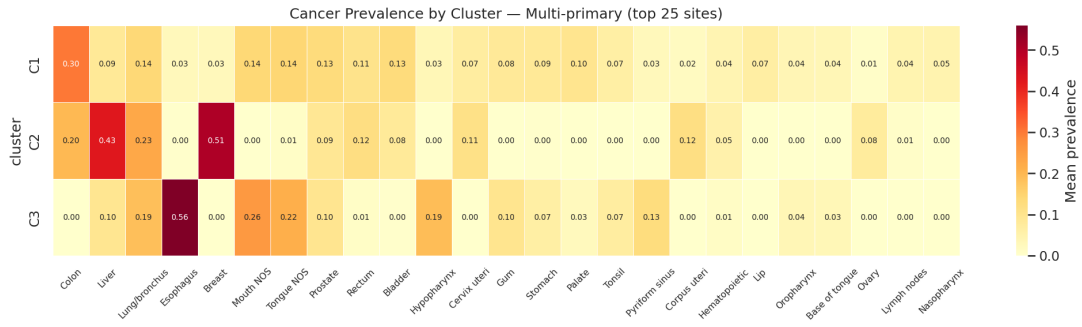


Figure 4: **Cancer-site prevalence by autoencoder cluster ($k = 3$).** Rows are ICD-O C-code sites; columns are the three clusters. Colour intensity encodes the fraction of cluster patients with a diagnosis at that site. Cluster C3 (upper aerodigestive SCC field) shows high prevalence of esophagus, hypopharynx, and oral cavity sites; C2 shows breast and gynaecological predominance; C1 shows diffuse GI/visceral involvement.

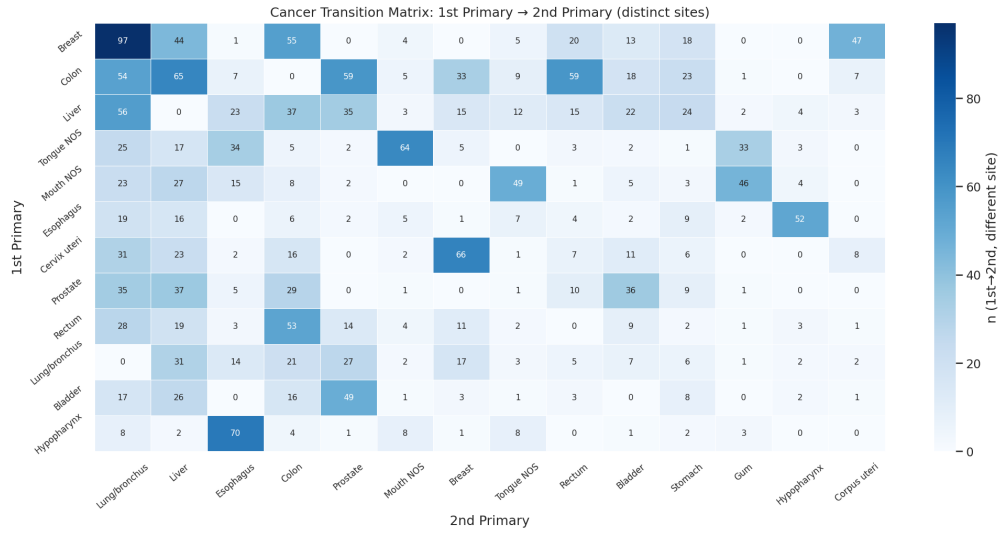


Figure 5: **First → second primary cancer transition matrix.** Each cell encodes the count of patients whose first diagnosed cancer (row) was followed by a second at the listed site (column). The bidirectional intensity within the upper aerodigestive block (esophagus–hypopharynx–tonsil, upper-left quadrant) contrasts with the predominantly unidirectional breast → lung and colon → liver transitions.

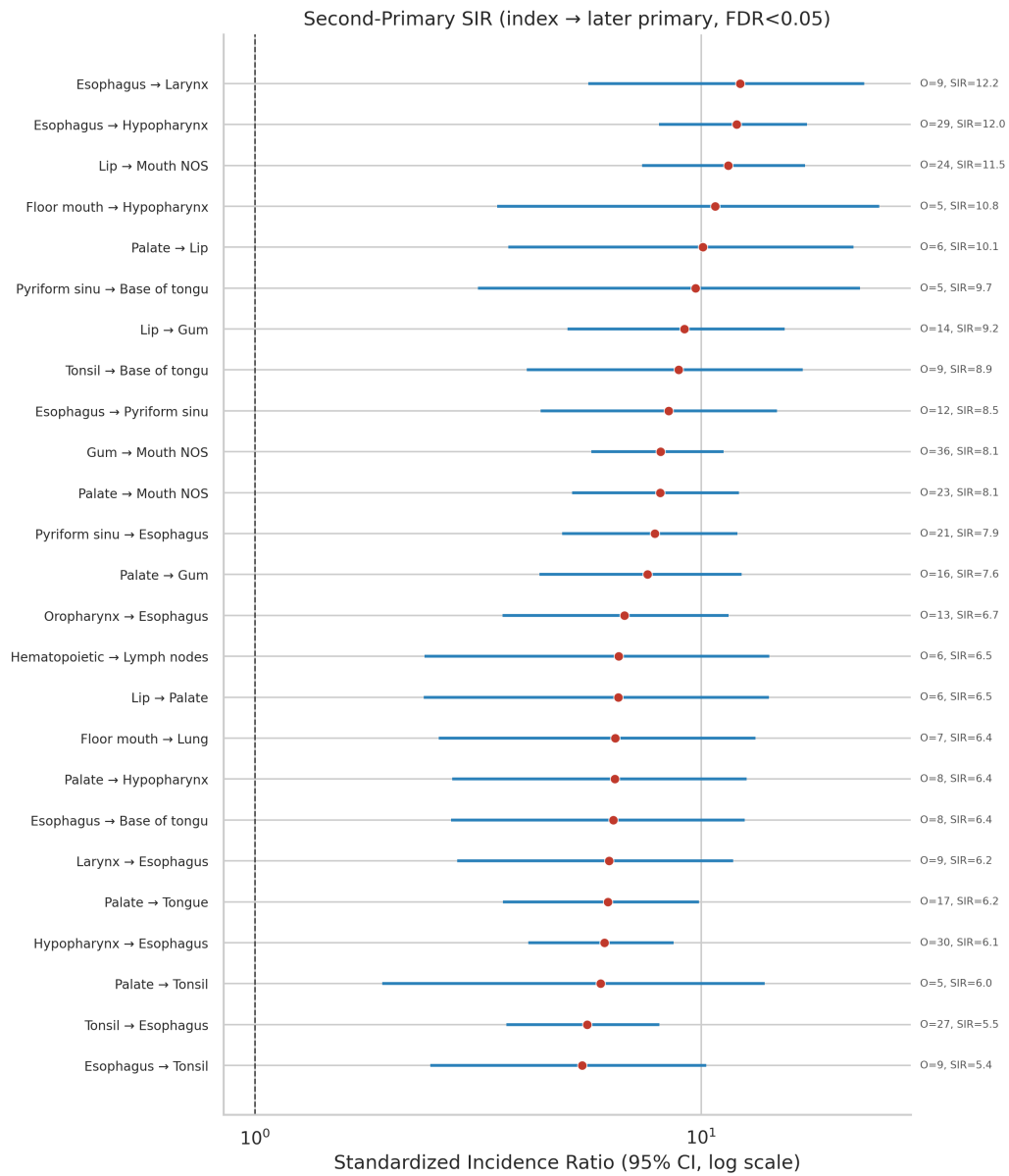


Figure 6: **Standardised incidence ratios for second primary cancers (FDR < 0.05).** Forest plot of SIR with 95 % Poisson confidence intervals for the 69 index → target pairs that reached FDR < 0.05 after Benjamini–Hochberg correction. The highest SIR values are confined to the upper aerodigestive field: esophagus → larynx (SIR 12.2) and esophagus → hypopharynx (SIR 12.0). Dashed line at SIR = 1 denotes no excess risk.

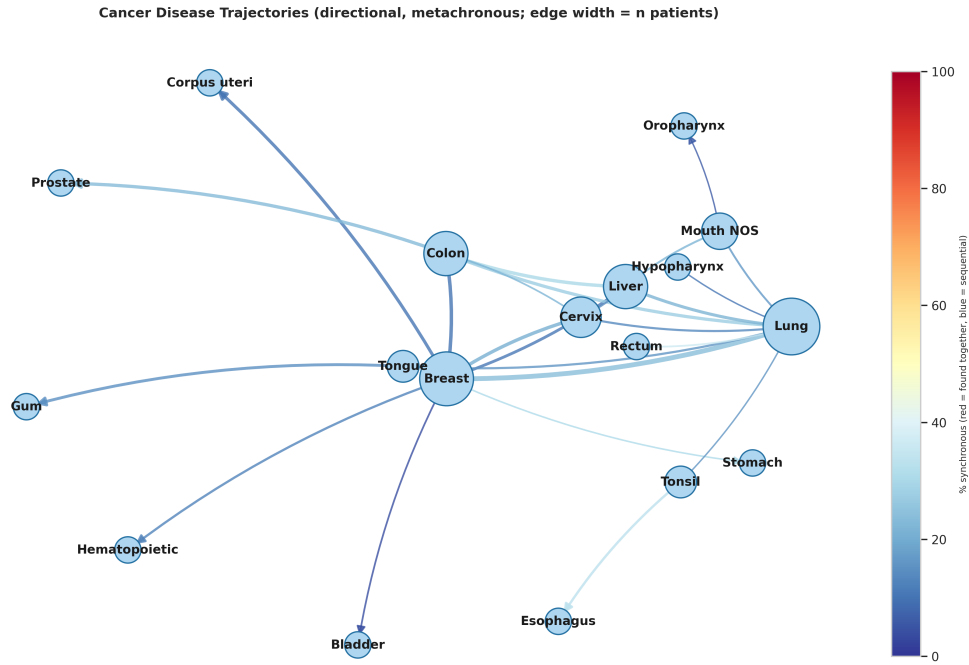


Figure 7: **Directional disease-trajectory graph.** Nodes represent cancer sites; directed edges denote significantly directional second-primary sequences (binomial FDR < 0.05). Edge colour encodes the percentage of pairs classified as synchronous (≤ 6 months); edge weight encodes pair count. Aerodigestive pairs (upper cluster) are bidirectional (grey, non-significant direction) with predominantly synchronous presentation, whereas breast and colorectal second primaries (lower-right) show strong unidirectional, metachronous flow.

Table 1: NMF cancer program clinical profiles (multiple-primary cohort, $n = 4,068$). Programs are ordered by male percentage.

Program	Theme	n	Male %	Median age	≥ 3 cancers %
P2	Upper aerodigestive SCC	615	97	53	17
P5	Oral cavity field	758	95	52	19
P7	Urological	406	90	72	5
P3	Lung-dominant	483	74	64	8
P4	Liver/GI	766	69	64	7
P1	Colorectal	310	57	64	2
P6	Female genital/breast*	730	5	56	3

Median age = median age at first diagnosis (years). *P6: Male % = 5 (i.e., 95 % female). Mortality % from registry vital status at last contact.

Table 2: Autoencoder cluster clinical profiles ($k = 3$, $n = 4,068$ multiple-primary patients). Clusters are labelled by dominant cancer domain.

Cluster	Domain	n	Median age	Male %	Mortality %	≥ 3 cancers %
C3	Aerodigestive SCC field	1,071	55	96	68.8	13
C1	GI/visceral mixed	1,827	61	76	58.8	11
C2	Female-enriched/breast	1,170	61	30	49.1	4

Mortality % = proportion dead at last registry contact (not actuarial survival).

Table 3: Disease-trajectory directionality and synchronous/metachronous split for selected cancer pairs. Forward and Reverse counts reflect metachronous directional pairs; % sync = proportion of all pairs (including synchronous) diagnosed within 6 months.

Trajectory pair	Forward	Reverse	% sync	Temporal mode	Dir. FDR
Hypopharynx $\leftrightarrow$ Esophagus	33	29	60	Field — synchronous	ns
Esophagus $\leftrightarrow$ Larynx	12	9	46	Field — mixed	ns
Breast $\rightarrow$ Lung	90	10	24	Sequential	<0.001
Breast $\rightarrow$ Corpus uteri	51	10	10	Sequential	<0.001
Colon $\rightarrow$ Liver [†]	57	22	30	Sequential	0.002
Cervix $\rightarrow$ Breast	57	22	21	Sequential	0.002
Colon $\rightarrow$ Prostate	55	21	23	Sequential	0.002

Dir. FDR: Benjamini–Hochberg-corrected binomial directional test; ns = FDR ≥ 0.05 (not significantly directional). Forward/Reverse: counts of ordered metachronous pairs; % sync includes synchronous pairs in denominator. [†]The colon $\rightarrow$ liver trajectory may include colorectal liver metastases coded as synchronous primaries; sensitivity analysis excluding this pair did not alter conclusions regarding upper aerodigestive field dominance.